

Annexure No.:

EXPERIMENT -05

Aim :

Introduction to arduino IDE with light blinking

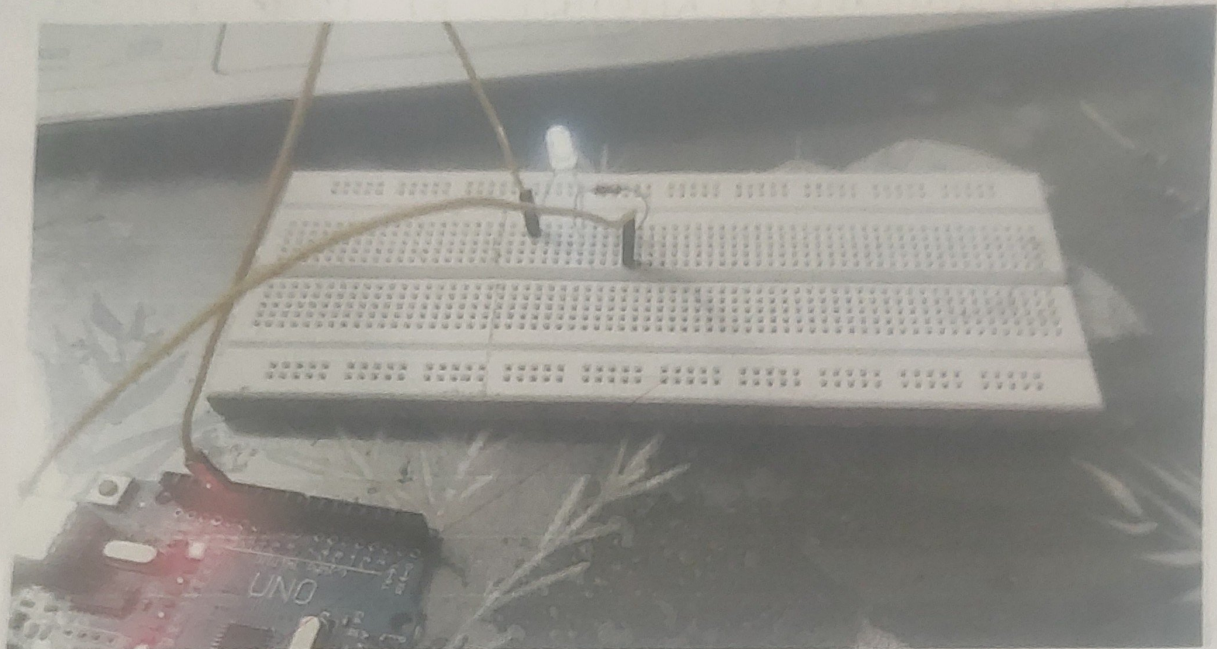
Apparatus :

LED bulb, bread board, wires resistors, USB cable, Arduino UNO.

Theory :

- Arduino IDE is an open-source software, designed by Arduino.cc and mainly used for writing, compiling & uploading code to almost all Arduino Modules.
- It is an official Arduino software, making code compilation too easy that even a common person with no prior technical knowledge can get their feet wet the learning process.
- It is available for all operationg systems ie. MAC, windows, linux and runs on the java plat form that comes with inbuilt functions and commands that play a vital role in debugging, editing and compiling the code.
- A range of Arduino modules available including Arduino Mega, Arduino UNO, Arduino Leonardo, Arduino Micro and many more.
- Each of them contains a microcontroller on the board that is actually programmed and accepts the information in the form of code.

diagram / performance OUTPUT :



CODE :

```
int ledPin = 13;
void setup () {
    pinMode (ledPin, OUTPUT);
    Serial. begin (9600) ;
}
void loop () {
    digitalWrite (ledPin, HIGH);
    Serial.println ("LED ON");
    delay (5000);
    digitalWrite (ledPin, LOW);
    Serial.println ("LED OFF");
    delay (1000);
}
```

Procedure :

- ① build circuit: plug on LED into a breadboard connect a 220Ω resistor from anode (+) to digital pin 13 and the cathode (-) to GND.
- ② connect Hardware: plug the arduino into your PC via USB.
- ③ Configure IDE: open Arduino IDE, under tools, select board and your COM PORT.
- ④ Write code: open Arduino IDE, under in setup () and delay () in loop ().
- ⑤ upload : click the verify (checkmark) then upload (arrow) icons.

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conclusion :

Successfully controlled on LED by programming on arduino pin as a digital output [VCC this confirmed the integration at software logic with hardware to perform outmated tasks.

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